ELLINGTON CYBER ACADEMY

Simulation 1 · Investigation Results

Finding security events in Jordan’s billing activity

Learner

hkuti

Case scope

EHS-SIA-SIM-001 · 2026-10-01 · 14:00–14:15 UTC

Time-window instructions

Keep the focused case window for all searches.

Report generated (UTC)

2026-11-09T01:19:35.236Z

Knowledge-check result

6/6 · 100.0% · Passed
Attempt: 1
Attempt submitted (UTC): 2026-11-09T01:19:12.036Z
Best score: 6/6

Assessment scope

Passing threshold: 80%. This is a browser-generated practice result, not an instructor-verified assessment. Written findings are not automatically graded.

Investigation findings

Which source IP associated with Jordan needs further investigation?

192.0.2.44

How many failed Jordan logins came from that source in the investigation window?

5

Which application is affected?

ehs_billing

Which target asset is affected?

EHS-WS-01

Which event records the successful login from that source?

login_attempt

How many failed attempts precede Maya’s internal success in this window?

2

Choose the disposition: Escalate, Close as benign, or Request more evidence.

Escalate

Choose the supported conclusion: Suspected unauthorized authentication, Confirmed data theft, or Confirmed account compromise.

Confirmed account compromise

Jordan’s supporting failure event IDs

evt-000001, evt-000002, evt-000003, evt-000004, evt-000005

Maya’s supporting event IDs

evt-000011, evt-000012

Jordan and Maya comparison

jblake's login attempts showed several logins from his expected private IP address (10.20.1.21) and a handful of failed login attempts from an unexpected public IP address (192.0.2.44) before an accepted one. Within that same 15 minute period, we can see that jblake had also had a successful login via his expected private IP address.

mthompson's login attempts showed only two failed attempts from her expected private IP address (10.20.1.31) before a successful one.

mthompson's login attempts were within a reasonable expectation of a potential typo or forgotten password while coming from an expected IP address while jblake's had a series of attempted logins from an unknown IP address before successfully logging in.

Decision reasoning

During the short 15 minute time period, we can see two successful logins from jblake's corporate-assigned private IP address as he presumably moves between pages and reauthenticates. Meanwhile, during the same 15 minute period, we can see an unknown public that repeatedly attempts to login to his account at the same time. There were 5 failed login attempts from this unknown IP address before a successful login.

This is evidence of a successful brute force attempt, and we can be extra sure of this because of the fact that because Jordan is using his expected private IP address to login at the same time. 5 failed login attempts from an unknown IP address is a red flag, while 1 or 2 can be a sign of a forgotten or mistyped password.

Remaining unknowns

We still don't know anything beyond the successful login attempts. We don't know if the attacker was able to access any other accounts beyond this, if they accessed any sensitive information or assets, if there was any malware dropped or involved, or if there was any data exfiltrated.

The information provided from this short snippet showed us the root cause; a successful brute force login from an unknown IP address. We need to further investigate what the full scope of the incident was, or if the attacker was successfully able to do more than simply login.

Additional handoff notes

I've pinned all events that originated from the unknown public IP address. This includes the 5 failed login attempts, as well as the single successful login attempt. These events happen alongside several successful login attempts originating from the expected private IP address. These all happened on Oct. 1, 2026 between 14:00:00.000 UTC and 14:15:00.000 UTC. There are several queries used to help confirm this evidence:
1. The full details of jblake's successful and failed login attempts:
scenario_id="EHS-SIA-SIM-001" user_id="jblake"
| sort(field=@timestamp, order=asc, limit=2000)

2. A simplified table view of jblake's successful and failed login attempts alongside the src_ip address for each one:
scenario_id="EHS-SIA-SIM-001" source_type="authentication" user_id="jblake"
| groupBy([user_id, src_ip, outcome], function=count())

Pinned evidence IDs

evt-000006, evt-000005, evt-000004, evt-000003, evt-000002, evt-000001

Knowledge-check answers and feedback

The case occurred on October 1, 2026, from 14:00–14:15 UTC. Which Falcon search setup best matches it?

Selected: Select the assigned training scope and that absolute UTC window.
Correct
Feedback: The assigned scope and absolute event-time window are needed to retrieve this historical case.
Correct response: Select the assigned training scope and that absolute UTC window.

In Jordan’s original authentication dataset, which field identifies the employee account?

Selected: user_id
Correct
Feedback: Jordan’s original records use user_id; newer datasets use user_name.
Correct response: user_id

Within 14:00–14:15 UTC, how many Jordan login failures came from 192.0.2.44?

Selected: 5
Correct
Feedback: The original records contain five matching failures, followed by a success from that account and source.
Correct response: 5

A success follows repeated failures from an unexpected source. Which conclusion is supported?

Selected: Suspected unauthorized authentication needs escalation.
Correct
Feedback: The pattern supports escalation for validation; it does not prove compromise or theft.
Correct response: Suspected unauthorized authentication needs escalation.

What does this CQL pipeline produce for the selected scope and time window?

Selected: A record count for each account, source, and outcome combination.
Correct
Feedback: groupBy creates groups from the listed fields; count counts the records in each group.
Correct response: A record count for each account, source, and outcome combination.

Which handoff best allows Riley to reproduce your authentication finding?

Selected: The scope, UTC window, exact query, event IDs, and evidence limits.
Correct
Feedback: Scope, time, query, and stable event references make the reasoning reviewable.
Correct response: The scope, UTC window, exact query, event IDs, and evidence limits.

Search queries

scope

```
scenario_id="EHS-SIA-SIM-001"
```

jordan

```
scenario_id="EHS-SIA-SIM-001" user_id="jblake"
| sort(field=@timestamp, order=asc, limit=2000)
```

summary

```
scenario_id="EHS-SIA-SIM-001" source_type="authentication" user_id="jblake"
| groupBy([user_id, src_ip, outcome], function=count())
```

maya

```
scenario_id="EHS-SIA-SIM-001" user_id="mthompson"
| sort(field=@timestamp, order=asc, limit=2000)
```

fields

```
scenario_id="EHS-SIA-SIM-001"
| sort(field=@timestamp, order=asc, limit=2000)
| select([event_id, @timestamp, user_id, src_ip, application, asset_id, action,
outcome])
```

Upload this report to the cyber range community. Exporting does not submit your work or update course completion.